



Intro to AI

Chapter 01:

A Universal field

it is relevant to any intellectual task

② Think humanly Computers think like humans	③ Think Rationally Study the mental faculties - Right decisions
① Acting humanly machines perform functions that require people intelligence	④ Acting Rationally Concerned with intelligent behaviour - affects intelligent agents.

① Acting humanly: Turing (1950) "Computing machinery and Intelligence"
"Can machines behave intelligently?"

Operational Test (the imitation game)

Prediction: - By 2000, a lay person can be fooled for 5 mins (30%)

- Suggested Major components of AI: Knowledge, reasoning, language learning, understanding
- Problem: the Turing test is not reproducible, constructive or amenable to mathematical analysis

Requirements to pass the test:

1. Natural Language Processing: (communicate successfully in a language) "Eng"
2. Knowledge Representation: (to store what it knows/hear)
3. Automated Reasoning: Use what's stored to draw conclusions.
4. Machine learning: adapt to new circumstances and to detect and extrapolate patterns.

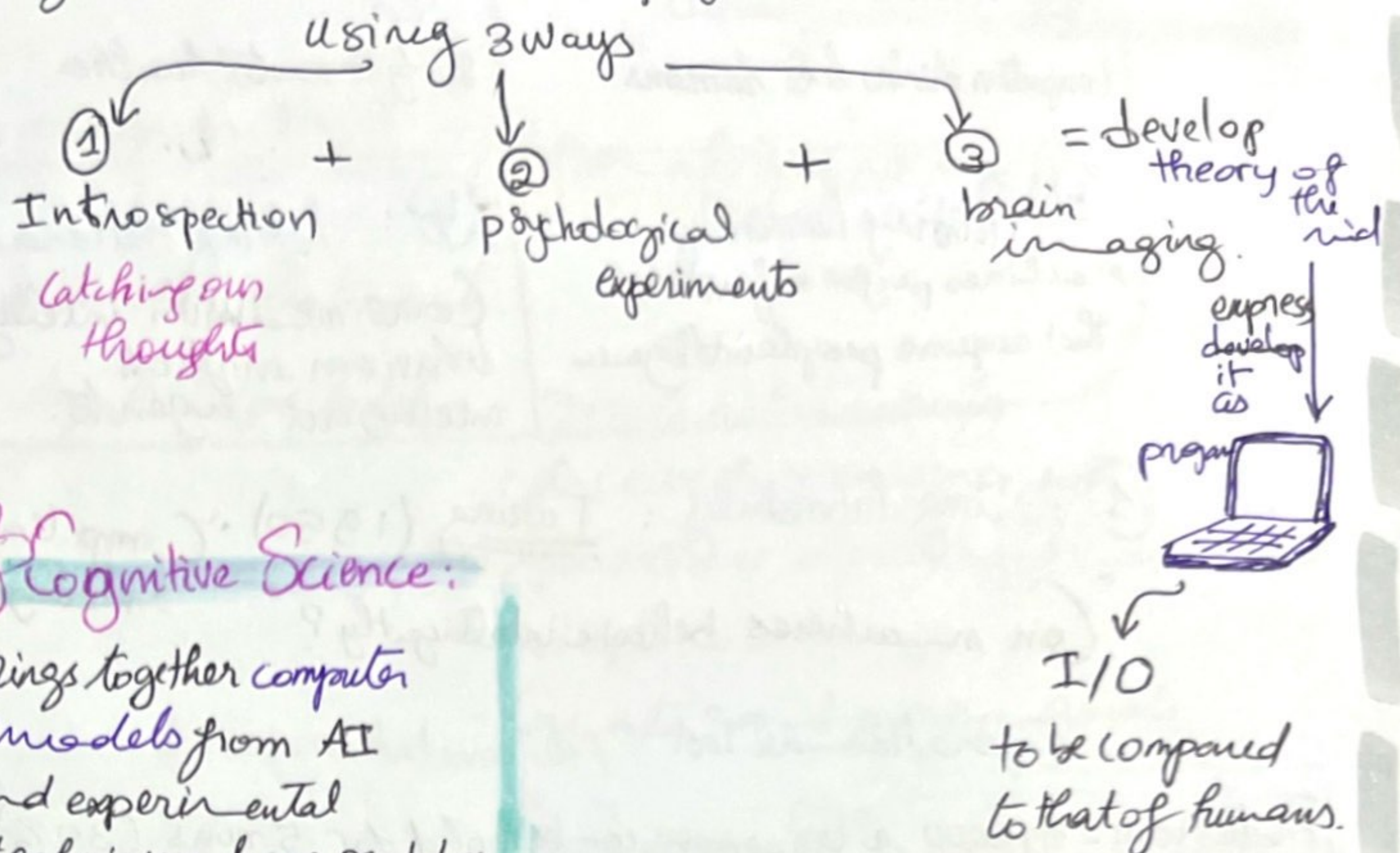
and • video signal: test the subject's perceptual abilities.

- opportunity for the interrogator to pass physical objects through the opening → needs Robotics manipulate objects

6. Computer Vision to perceive objects

② Thinking humanly: Cognitive modeling approach.

- Determine How humans think.
get inside the actual workings of human minds.



Cognitive Science:

brings together computer models from AI and experimental techniques from psychology to construct precise and testable theories of the human mind.

How human mind functions

③ Thinking Rationally: Laws of thought approach initiated the field called logic

→ Aristotle is the first to attempt to codify the right thinking

→ His Syllogisms provided patterns for argument structures that always yielded correct conclusions for correct premises

logician → build on intelligent systems.

2 main obstacles: ① Not easy to take informal knowledge and state it in the formal terms required by logical notation particularly when knowledge is not 100% certain

② Complexity: even with only hundred facts, it can exhaust the computational resources.

④ Acting Rationally: "Rational Agent" Approach

★ Agent: is an entity that acts

★ Computer Agents: expected to:

- operate autonomously
- perceive their environment
- persist over a prolonged time period.
- Adapt to change
- Create and pursue goals.

Rational AGENT: The one that
Acts to achieve the best Outcome
or when there is uncertainty → best expected outcome

• Correct ^{inference} is not all of rationality
→ in some situations there is no provably correct thing to do but something must still be done.

→ WAYS of acting rationally without involving any inference like REFLEXES (blinking the eye).

It's the best thing to do but nothing to conclude

→ skills needed for the turing test = allow agent act rationally.

Which will
maximize
goal achievement
given the
available
information

2 Advantages of Rational Agent approach over other approaches:

- 1- MORE General than the Laws of thought
- 2- MORE AMENABLE to scientific dev. than approaches based on human behaviour/thoughts.
- Rationality is mathematically well defined. and can lead to agent designs that are provably achieve it!

Foundations of AI

→ It has emerged from research in various fields

① Philosophy: reasoning

- Aristotle → developed informal system of syllogisms for proper reasoning. [allowed one to generate conclusions mechanically given initial premises].

Syllogism: uses two premises to draw a conclusion:

for example:

- Major premise: general statement about a broad category
- Minor premise: specific statement about a subset of the category.

→ Conclusion: drawn from the relationship between them.

Example: Major: Men are mortal

Minor: Socrates is a man

Conclusion: Socrates is mortal.

René Descartes (1596 - 1650): first clear discussion of the distinction between mind and matter.

- If the mind is entirely governed by physical laws then it has no more free will than a rock falling on the ground.

→ Believes in the power of reasoning in understanding the world (rationalism) + also proponent of dualism.

- there is a part of the human mind (soul or spirit) that is outside of nature, exempt from physical laws.

≠ Alternative to dualism is materialism which holds that the brain's operation according to the laws of physics constitute the mind.

2 Source of Knowledge:

Empiricism: movement started with Francis Bacon

(Reference to observation, experimentation)

↓
Rationalism = rationalism + Empiricism
dualism
materialism

- Induction. With David Hume (1711 - 1776).
- Acquisition of knowledge from experience.

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Finish from the slides

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in
the
Chapter 2